

Lab 11: Printed Circuit Board

AE 8883 Spacecraft Engineering

Summary:

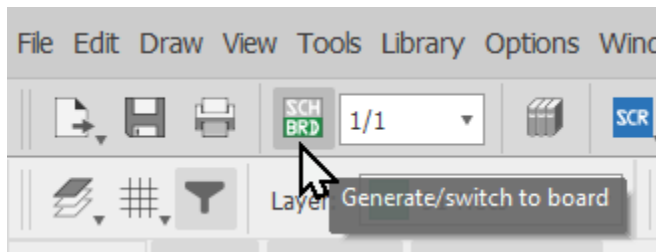
For this assignment, you will be taking the completed electrical schematic from the previous lab, and laying out its components on a PCB in Autodesk EAGLE. This will involve defining a size for your PCB, arranging components on it, and routing traces between components. At the end, a DRC will be performed to verify your board is up to the manufacturing standards of a PCB fabrication house.

Application:

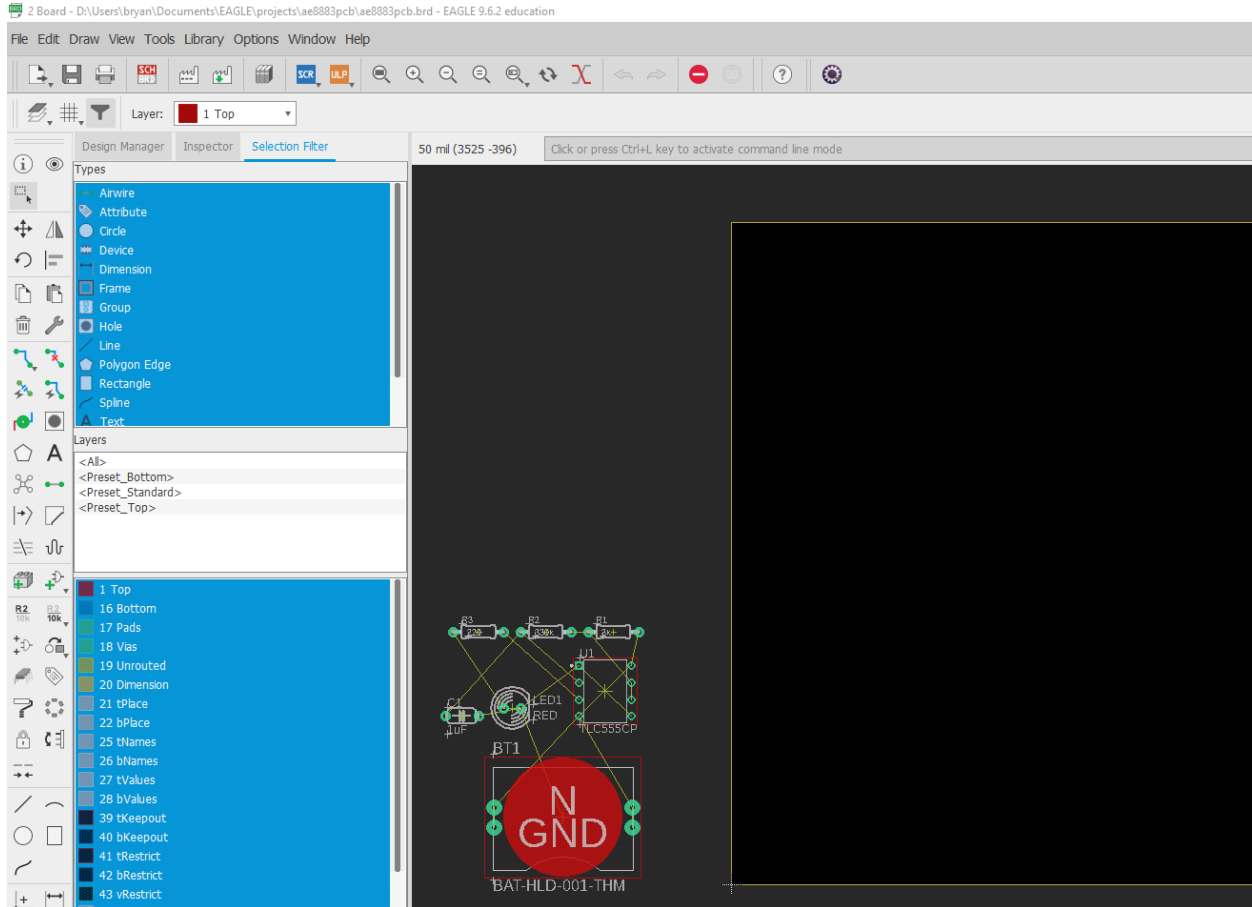
EAGLE Board and DRC Setup

Upon starting EAGLE, you should have a Control Panel window open.

1. Make sure the **ae8883eagle.lbr** EAGLE component library from the previous lab is still being Used for the current lab.
 - a. If it isn't, tell EAGLE that you want to **Use** this library for projects you're working on this session by clicking on the gray dot to turn it green. Alternatively, you can right-click the library and click **Use** to check it and achieve the same result. In EAGLE, all imported libraries must be included in projects in this matter for them to show up in the part selection menu.
2. In the **Control Panel**, expand the **Projects > projects > ae8883pcb** project you created last time. Make sure you are currently Using the project (make sure the gray circle is green) and reopen your schematic.
3. Click the Schematic/Board button in the toolbar to switch to **Board** view. If you don't currently have a board view associated with your schematic, let EAGLE create one for you from the schematic.



A Board view window should now pop up that looks something like the following:



We will now load a DRC (Design Rule Check) file to automatically have EAGLE constrain design choices based on the manufacturing capabilities of our chosen PCB fabrication house. In this case, we'll use OSH Park's 2 layer board DRC file.

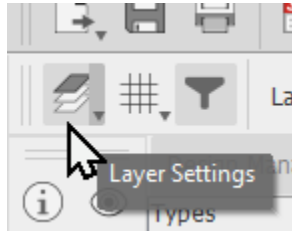
1. If you haven't already, download the provided **oshpark-2layer.dru** EAGLE DRC file that accompanies this lab. It can be downloaded from the Printed Circuit Board Canvas Module
2. In the **Board** window, select **Tools > DRC**
3. In the File tab, select **Load...** and open up the **oshpark-2layer.dru** file
4. Hit **Apply** and then close out of the DRC window. We'll come back to it after our layout is finished to perform the check.

You'll find a **Grid** button at the top left that allows up to set the size of the grid your components will snap to. Units can be set to mils (1/1000 of an inch) or other units such as mm (millimeters). For now, we'll stick with mils. You can choose to display this grid if you wish by checking the respective selection

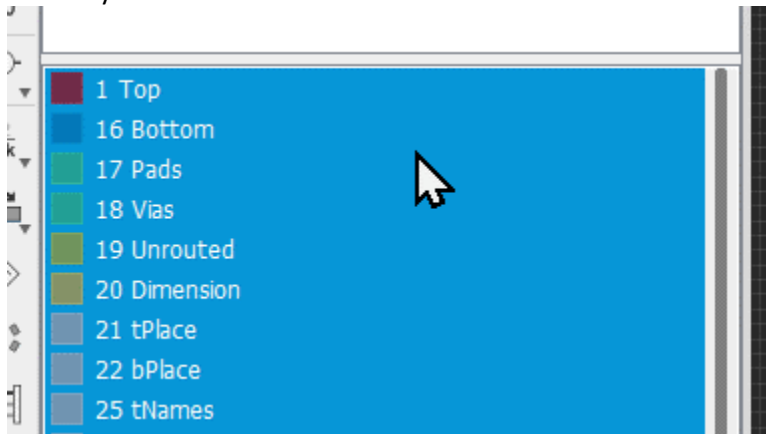
button.



The button to the left of the Grid button can be used to toggle the visibility of layers on and off.

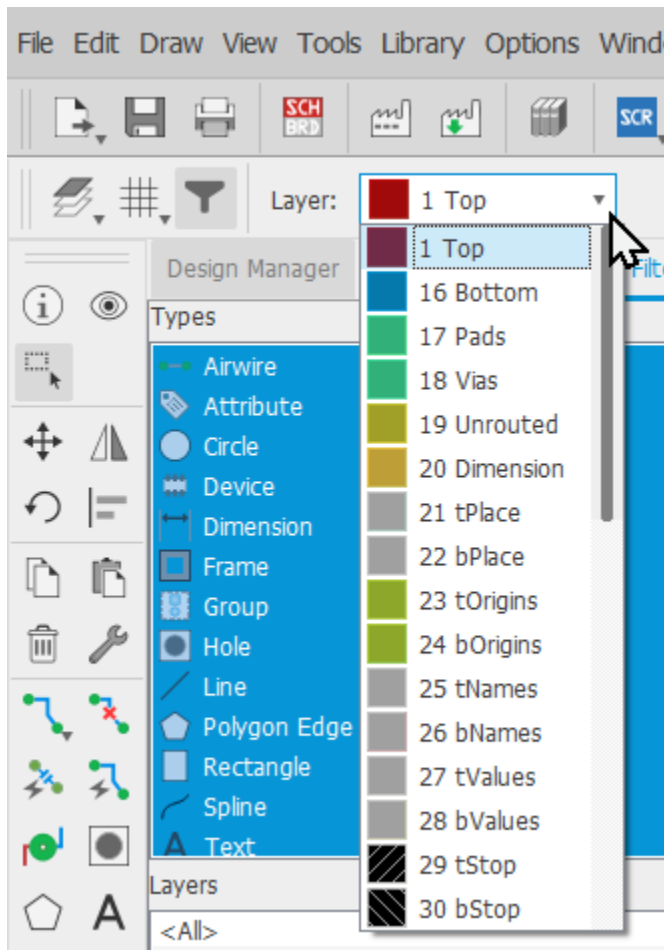


The layer menu at the bottom of the sidebar, on the other hand, dictates which layers are selectable/interactable in the viewport. You may find this handy if you ever need to work with dense, multi-layered PCBs.



EAGLE Board Layout

At the top right, you'll see a Layer dropdown that lists all the layers you can interact with on this PCB.



Not all layers are actual physical layers on the PCB, but they're useful for organization. The lowercase **t** and **b** prefixes in front of some names signify whether that layer is used for the top or the bottom of the board.

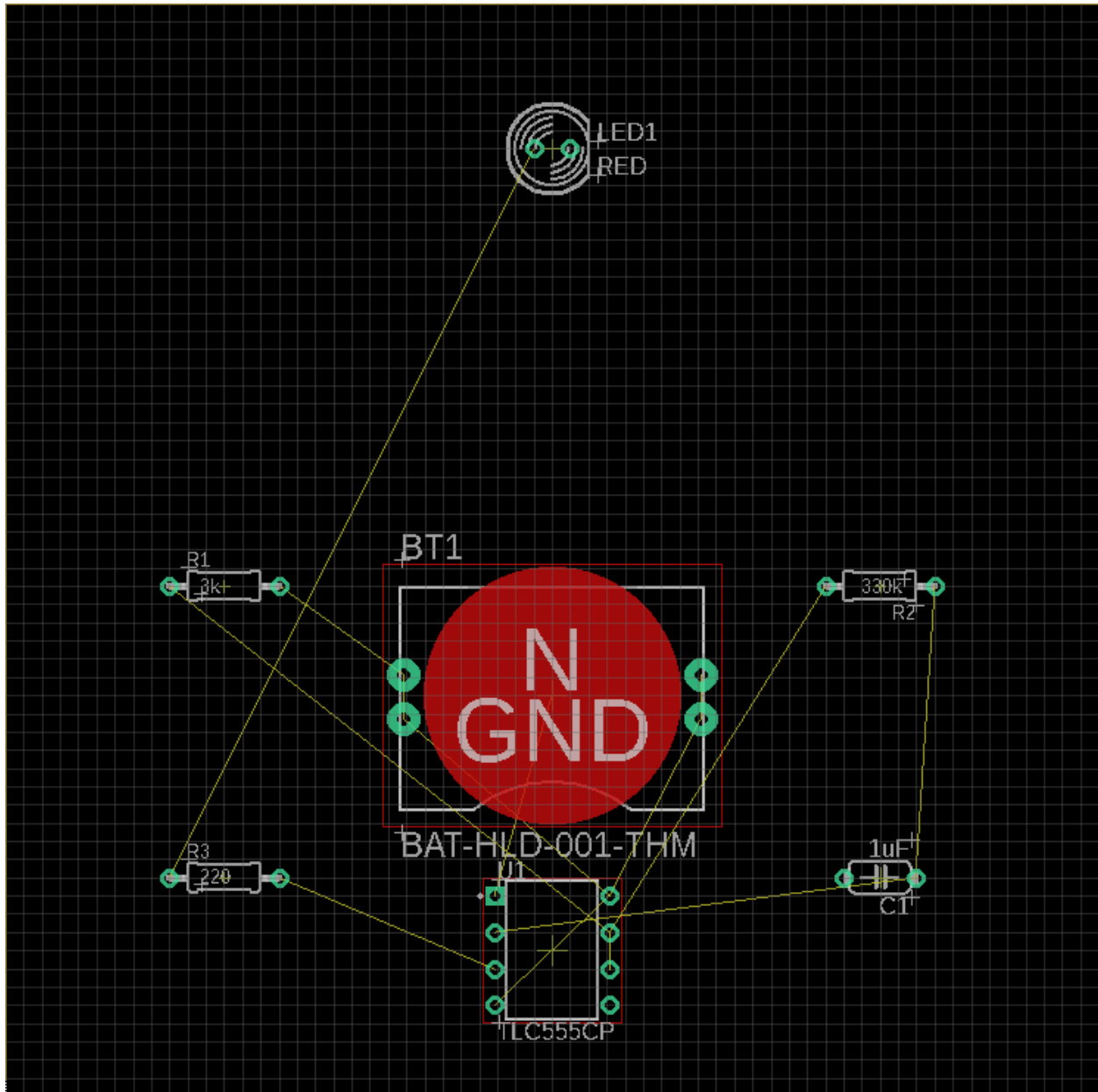
We'll first change the size of our PCB. This is defined by the yellow line in the **Dimension** layer.

1. (optional) Use the **Layer** dropdown to select the **20 Dimension** layer as the currently active layer. Entering **layer 20** in the command line will also achieve the same effect.
2. Right click the lines of this yellow box and select **Properties**. Change the coordinates of the sides of the board to turn it into a 3-inch by 3-inch square (3000 mils by 3000 mils).
3. If you want, you can draw more exotic shapes to define the borders of your PCB. Just make sure you're doing so on the Dimension layer, or move those lines to the Dimension layer afterwards.

Now you're free to arrange the components on this square however you like. Just make sure you keep everything within the borders of the yellow square.

- Group (used for selection), Move, and Rotate tools can be found on the left toolbar. Right-clicking while a part is being dragged around will also rotate the part.

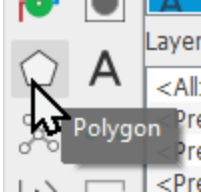
- The Mirror tool will mirror the affected component. What this means is that it will be transferred to the opposite side of the board it is currently on. You'll notice this has occurred if the color of the pads of your component turn from red to blue. In EAGLE, these are color-coded to the top and bottom of the PCB respectively. Middle-clicking while a component is being dragged around will achieve the same effect. For the purposes of this lab, it should be easier to keep everything on one side of the board, so just leave everything on the top layer.



You'll notice that there are thin yellow lines going between the pins of the different components you have. These are called **airwires**. EAGLE displays these on-screen to let you know that your schematic says there should be an electrical connection between these two points, but it doesn't see one drawn on the board. To form these connections, we'll connect these points using traces in a bit.

Before that though, we know from the start that we want our board to have a ground plane. This will be the bottom layer of the PCB, color-coded blue. Let's first draw a polygon to mark this plane out.

1. Use the **Layer** dropdown to select the **2 Bottom** layer as the currently active layer. Entering **layer 2** in the command line will also achieve the same effect.
2. Select the **Polygon** tool in the left sidebar. Or type **polygon** in the command line.

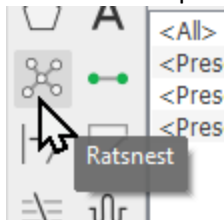


3. Use the tool to draw a polygon around the entire board boundary. Make sure it completely surrounds the yellow border.
4. Upon completing the polygon, give the polygon the name GND to let EAGLE know you want it to define a ground pour.
5. The polygon should now be a dotted blue line.

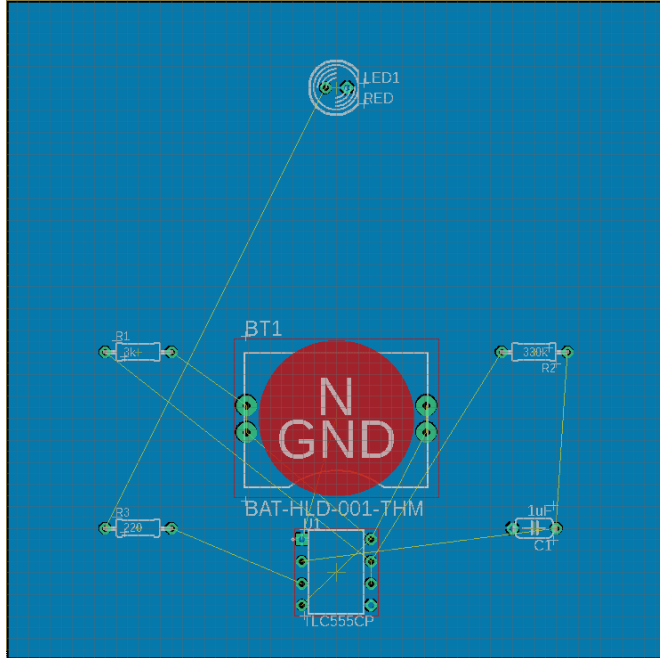
Now is a good time to try out the Ratsnest tool. In EAGLE it does the following things:

- Assesses existing airwires and looks for shortest possible paths according to your current component placement
- Calculates all polygons that belong to a signal

Go ahead and press it. Or type **ratsnest** in the command line.



Your board should now look something like the following (note that I switched back to the top layer for this screenshot):

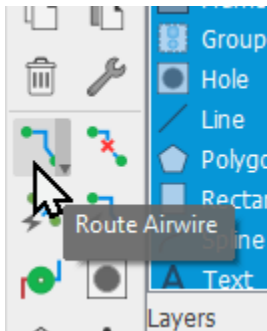


This blue can be kind of distracting though, so for now, let's hide the real pour and switch back to outline mode. To do this, type the following in the command line:

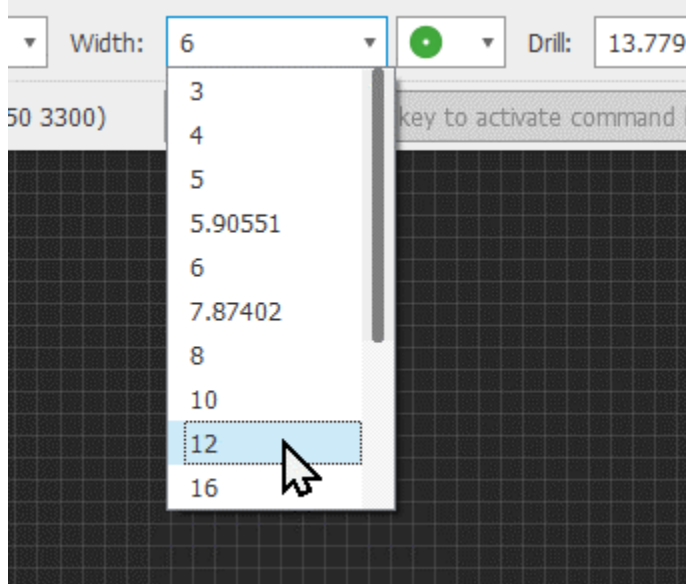
ripup @;

For now, we can save the routing of any GND signals for last, because we know we can just route those signals to the GND plane later. We'll instead focus on drawing traces between the remaining signals on the board. **Ignore any GND nets for now.**

1. Select the **Route Airwire** tool in the left sidebar. Or type **route** into the command line.

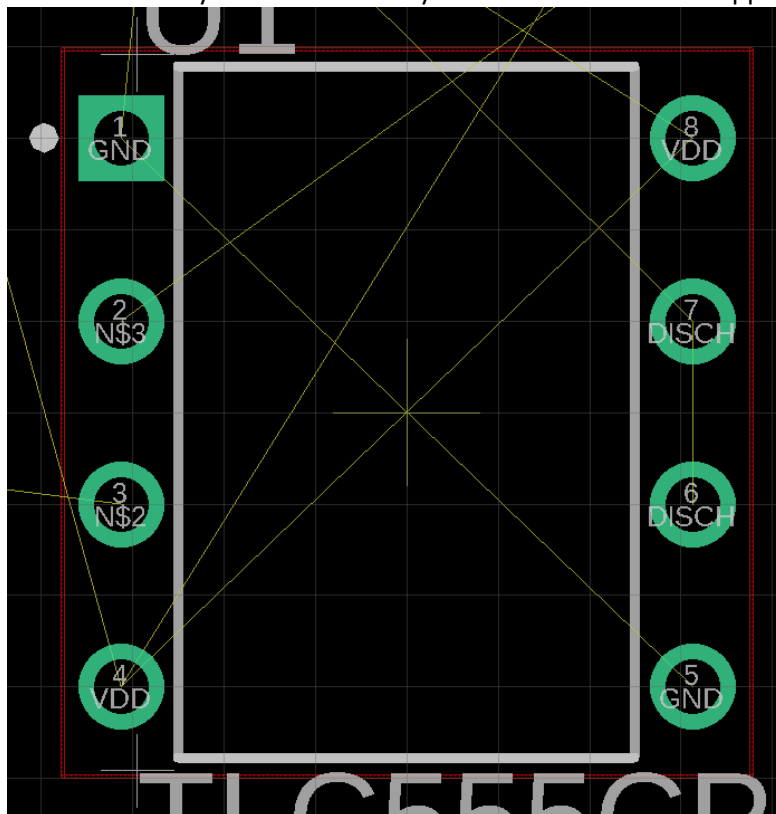


- When you've done so, you'll see additional options appear at the top. One of the dropdowns controls the Width of the trace you want to draw. Set this to **12 mils**.



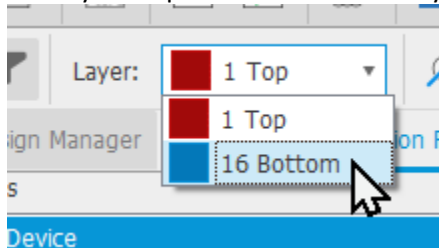
You can now use this tool to begin drawing traces between pins marked by the airwire. Note that you do not need to draw traces according to the airwires EAGLE displays for you, especially if a single net connects multiple pins together.

If you zoom into a pin, you can see the signal that pin carries written in gray. You can use this to determine how you want to route your traces between the appropriate signals.

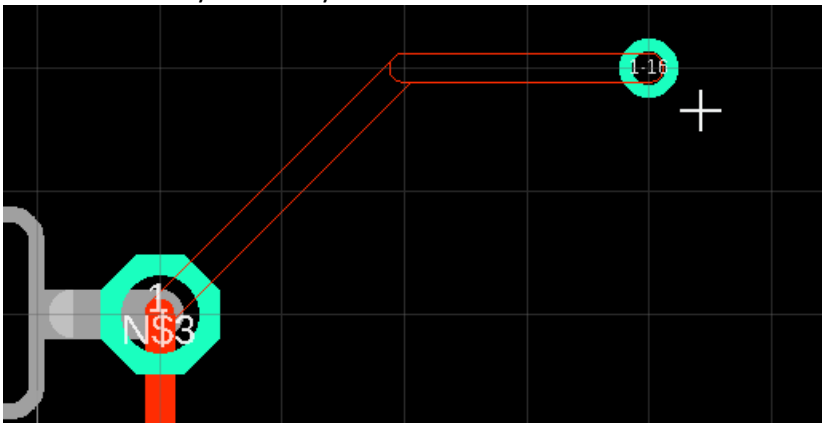


You may notice while routing your traces that there is no direct path to the pin you want to route to. You would need to go under an existing trace in order to get to the other side. To do this, we can add a via as part of our trace path.

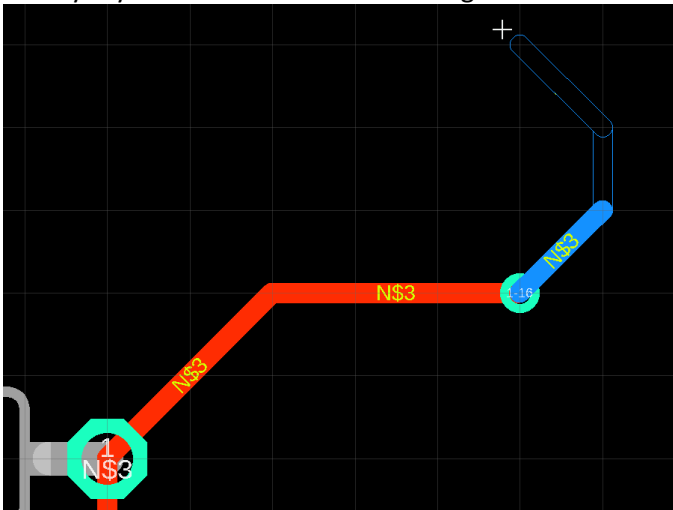
1. While the **Route Airwire** tool is active, and you are actively drawing a trace on the board, use the Layer dropdown to select the layer you want to change your current signal to.



2. You should now see a via at the end of the trace you're currently drawing. It will follow your trace around as you move your cursor.



3. Place the via down in your chosen spot. You'll now notice the color of the trace has changed to the layer you wanted to transfer the signal to.

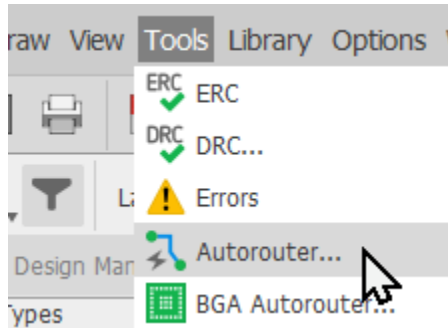


4. Route your trace as needed to its final destination, using vias to change layers as needed.

Remember that through-hole components are already connected to both layers through their vias, so you can connect to either side to form a connection. SMD pads however can only be connected to from the side they're on.

If you don't feel like routing traces, or have given up trying to do so, you can try out EAGLE's autorouter if you want. To do so:

1. Rerun Ratsnest to regenerate shortest paths.
2. Save your board.
3. Select **Tools > Autorouter...**

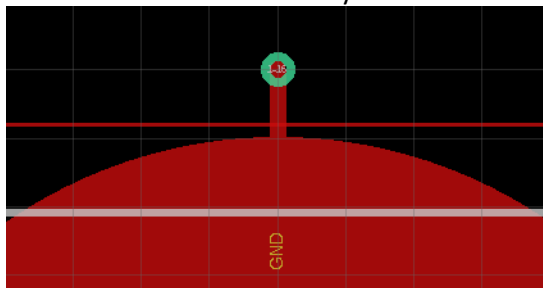


4. You'll see a menu pop up with some customization options. Preferred directions indicates how that layer is going to be used during autorouting calculations. A dash means horizontal routing is preferred (E.g. staying on the same layer) while a vertical bar means vertical routing is preferred (E.g. vias and signal transfer between layers). Effort and thread count can be adjusted based on the computing power of your workstation. CTL files can also be loaded for more configuration options.
5. Press **Continue** to confirm your choices, then hit **Start** on the next window that pops up.
6. If you want to keep the autorouted traces, you can hit **Evaluate** to save them.

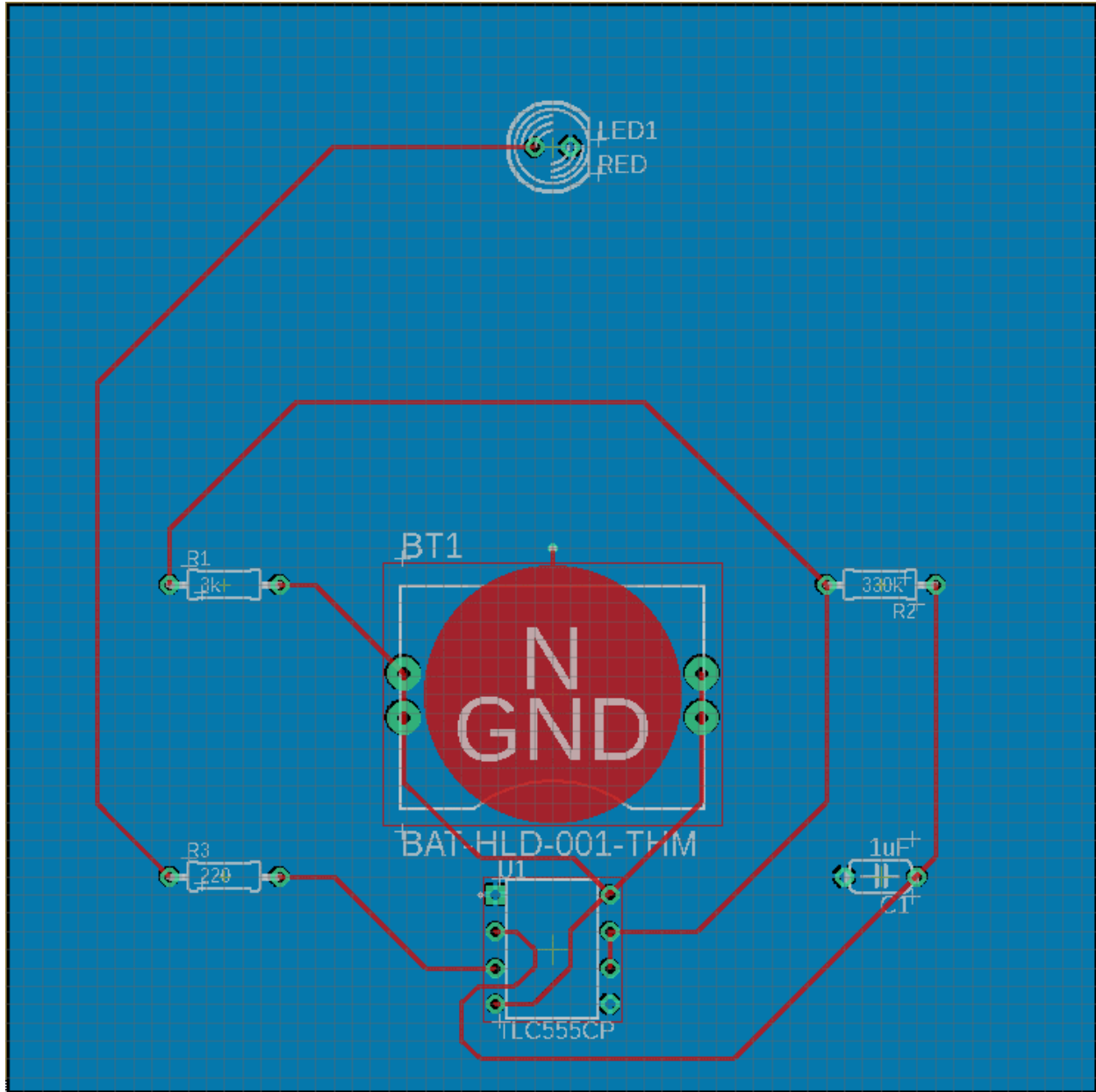
The end result is an acceptable circuit that routes traces using shortest path algorithms. Without extra configuration though, the end product is usually not the best, which is why we prefer to route traces on our own when possible.

For the purposes of this lab, you can either use the autorouter, or do it yourself. This lab will be proceeding with a manual routing.

The giant GND pad of our button battery holder is the only thing left to route, and we need to connect it to GND. Since the bottom layer of our board is GND, we'll just extend a via from it to do so:

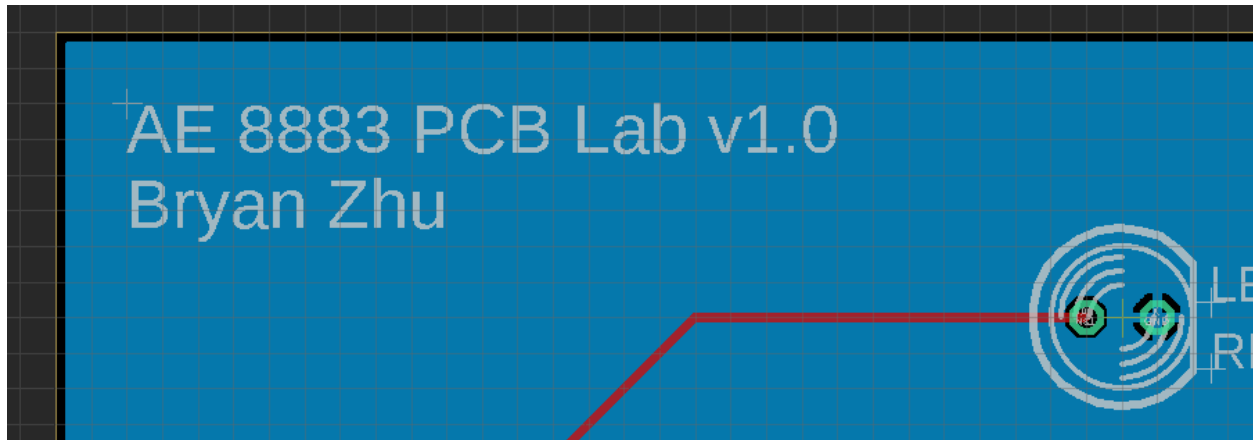


Now that our only pad is taken care of, we can hit Ratsnest one more time to resolve the remaining GND pins and tie them to our ground plane. This is our finished board:



The last thing we'll do is add some silkscreen text. This is desirable for documentation purposes.

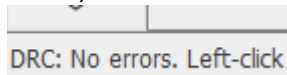
1. Use the **Layer** dropdown to select the **21 tPlace** layer as the currently active layer. Entering **layer 21** in the command line will also achieve the same effect. In EAGLE, tPlace and bPlace are the layers reserved for silkscreens on the top and bottom of the board respectfully.
2. Select the Text tool in the left sidebar.
3. Type out the following information into the text box. Use multiple lines if desired.
 - Project/Board Name
 - Designer Name
 - Board Revision Number
4. Place the silkscreen text onto the board.



If you want to add other silkscreen, feel free.

Great, we're done! Now let's run a DRC of our board. Go to **Tools > DRC** and hit the **Check** button.

DRC progress will be displayed in the status bar at the bottom left of the screen. If you don't get any errors, the status bar will let you know:



If you do get errors, address them as needed.

Once your board is error-free, you're done with the lab!

Take a screenshot of your final board or export an image using File > Export > Image as with the schematic. Include it in journal.